

Zhu Jingrui

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EDUCATION BACKGROUND

Sichuan University

Sep. 2023 – Jul. 2027

Bachelor of Engineering in Flight Vehicle Control and Information Engineering

GPA: 3.57/4.0 (Weighted Percentage: 86.23/100)

IELTS: 6.5 (Reading: 8.0) | CET-6: 583 | CET-4: 594

Relevant Courses: Automatic Control Principles, Modern Control Theory, Signals and Systems, Aerodynamics, Flight Dynamics, Digital Circuits, Analog Circuits

RESEARCH EXPERIENCE

Array Signal Processing and Collaborative Localization for UAV Swarms

Communication Algorithm Lab, Sichuan University | Core Member | Sep. 2025 – Jun. 2026

- Independently reproduced GCA-MUSIC and GCA-RMUSIC algorithms, achieving $<0.1\%$ relative error in DOA estimation under low SNR; completed the full pipeline from paper equations to executable code
- Participated in a national innovation project on TDOA/FDOA-based distributed UAV swarm localization; addressed the idealized assumption of precisely known reference station positions by introducing a position error vector into the observation model, reformulating it as a joint estimation problem
- Assisted in building a communication-constrained observation geometry model for multi-agent systems and explored low-complexity collaborative tracking strategies

PROJECT & INTERSHIP EXPERIENCE

Institute of Software, Chinese Academy of Sciences | Research Intern

Jul. 2026 – Sep. 2026

- Investigated intrinsic mechanisms of forearm pronation/supination illusion evoked by EMS stimulation for embodied intelligence and human-robot collaborative interaction
- Developed an experimental interaction platform integrating QTM optical motion-capture and Unity engine in C#; completed device calibration and platform debugging for human-subject experiments
- Organized human-subject trials, collected multi-source kinematics and subjective perceptual data; conducted data cleaning, statistical analysis and visualization with R and SPSS
- Compiled experimental results and analytical figures; co-authored a manuscript submitted to CHI 2027

Aerospace Xintong Technology Co., Ltd. | Intern Engineer

Jan. 2026 – Mar. 2026

- Conducted technical research on Tianmu satellite occultation data inversion, supporting product discussions and project meeting preparation
- Assisted the "PolarNet" team at Southwest University in researching electromagnetic feature characterization of low-orbit satellites and compiling project application materials

HONORS & AWARDS

First-Class Scholarship, Sichuan University | 2025

Second-Class Scholarship, Sichuan University | 2024

Outstanding Student, Sichuan University | 2024

SKILLS

- Python:** Proficient in data analysis and deep-learning development; familiar with CNN, RNN and Transformer architectures, with practical experience in PyTorch and NumPy
- MATLAB:** Experienced in matrix computation and algorithm simulation; familiar with Simulink modeling for dynamic system validation
- C:** Familiar with fundamental programming; basic practice in embedded development based on Keil for microcontroller applications
- Verilog:** Competent in FPGA-oriented digital system design, covering RTL coding to onboard verification
- R & SPSS:** Proficient in statistical analysis, repeated-measures ANOVA and experimental data visualization
- C# & Unity:** Experienced in developing experimental testbed integrated with QTM optical motion-capture system